Ogbonna Arinze

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SUMMARY

Full-stack engineer with 8 years of experience building high-impact web applications and production AI systems across healthcare, fintech, civic tech, and enterprise environments. Proven track record of owning products end-to-end, from architecture and database design through conversational AI pipelines and polished user interfaces using Next.js, Node.js, PostgreSQL, and LLM APIs. Built and shipped a multi-agent RAG platform from scratch as a solo engineer, and co-developed a conversational AI widget at Genentech (Roche). Thrives in ambiguous, fast-moving environments with a product-obsessed mindset, strong technical ownership, and experience mentoring engineering teams.

CORE COMPETENCIES

- **Full-Stack Development:** Next.js, React, NestJS, Node.js, TypeScript, REST API design, SSR/SSG, and monorepos (Nx, pnpm).
- **AI & Conversational Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), embeddings, prompt engineering, and multi-agent workflows.
- **Databases & Infrastructure:** PostgreSQL (pgvector), MongoDB, Prisma, Docker, GitHub Actions, CI/CD pipelines, AWS, and cloud deployment.
- **Frontend Architecture:** React, Vue.js, Redux, Zustand, Tailwind CSS, Radix UI, design systems, and performance optimization.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, ESLint, Prettier, and CI-driven QA.
- **Product & Leadership:** End-to-end product ownership, architecture reviews, mentorship, Agile/Scrum, and direct stakeholder collaboration.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer *May 2023 - Present*

- Co-developed a full-stack **Next.js chat widget** powered by a **Retrieval-Augmented Generation (RAG)** system using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query them in real time via a **conversational AI interface**.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, driving adoption improvements and streamlining clinical site integration workflows.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** in Next.js with TypeScript and React Query, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Optimized application performance by implementing **lazy loading, code splitting**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria

Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React, Redux, RxJS**, and **Axios**.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)

Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **Node.js RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, collaborating directly with the business team.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria

Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **Node.js**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **Node.js** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng

Senior Engineer *Feb 2026 - Present*

AI-powered platform for Nigerian budget analysis and corruption tracking. Multi-agent LLM pipeline analyzes 700+ budget documents (708K+ vector embeddings) across 37 states, enabling citizens to query government spending via conversational AI.

- Architected and built the entire platform from scratch as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with **pnpm** workspaces.
- Designed and implemented a **multi-agent AI system** using the **Mastra** framework with 5 specialized agents (Router, Budget Analyst, Corruption Analyst, GovSpend Analyst, Impact Analyst) for three-stage intent classification (keyword scoring, **LLM classification** with 5-min cache, inferred fallback) and domain-specific **RAG-powered** analysis.
- Built a **hybrid RAG pipeline** combining **pgvector** similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion (K=60), with **Cohere** rerank-2 reranking and **Voyage AI** voyage-3-large embeddings (1024-dim, 708K+ chunks across 4 vector indexes).
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, token-budgeted context management, conversation summarization, and memory extraction for a responsive conversational experience.
- Built the document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing 700+ documents (959 files) across 37 states with zero ingestion errors.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and wrote custom migration tooling for drift detection.
- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts**, featuring 22 interactive chart types auto-generated from AI responses, dark mode, URL state management, and responsive mobile layouts.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests (4 projects), and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and 30s graceful shutdown.

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*